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Disease Risks for Restoring Endangered Sahelo-Saharan Antelope: A Literature Review

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Abstract: In 1994, the Convention on the Conservation of Migratory Species of Wild Animals identified six species of Sahelo-Saharan antelope in immediate danger of extinction. Several of these species are now subjects of large-scale conservation efforts, including reintroductions and population reinforcements across their native ranges. However, disease-related mortalities can cause major setbacks, and wildlife-livestock interfaces are widely thought to have contributed to disease and mortality events in translocated animals. We reviewed 202 English and French publications spanning 20 years (2000–2020) and the WOAH-WAHIS database for infectious diseases and parasites that could pose risks to Sahelo-Saharan antelope. Our review included countries with active conservation translocation initiatives for these species: Chad, Morocco, Senegal, and Tunisia. We found reports of 159 distinct infectious agents or parasites. Frequently identified viruses included bluetongue virus, Rift Valley fever virus, and small ruminant morbillivirus. Commonly reported bacterial and protozoal

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agents included *Anaplasma spp.*, *Theileria spp.*, and *Toxoplasma spp.*, while common ectoparasites included *Hyalomma* and *Rhipicephalus* ticks. Sixty-nine (43%) of the infectious agents identified were potentially zoonotic, 39 (25%) were vector-borne, and 30 were WOAH-listed diseases. These findings highlight potential regional health threats to Sahelo-Saharan antelope and neighboring livestock and human communities. Targeted research is needed to elucidate the role of wildlife in the epidemiology of livestock diseases and vice versa.

Keywords: Sahelo-Saharan antelope, Livestock, Chad, Morocco, Senegal, Tunisia

Introduction

Many large vertebrates native to the Maghreb and Sahelo-Saharan regions, which stretch across more than 10 million km^2 of Central, North, and West Africa, experienced dramatic declines during the nineteenth and twentieth centuries (Brito et al., 2013; Durant et al., 2013; Newby et al., 2016). Five endemic antelope species—addax (*Addax nasomaculatus*), Cuvier's gazelle (*Gazella cuvieri),* dama gazelle (*Nanger dama*), scimitar-horned oryx (*Oryx dammah*, hereafter "oryx") and slender-horned gazelle (*Gazella leptoceros*)—are currently considered in danger of extinction. All five species are listed on Appendix I of the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) and Appendix I of the Convention on the Conservation of Migratory Species of Wild Animals (CMS), are target species of the Sahelo-Saharan Megafauna Concerted Action (Beudels-Jamar, 1998), and are classified as Critically Endangered (addax and dama gazelle; IUCN SSC ASG, 2016a, 2016d), Endangered (slender-horned gazelle and oryx; IUCN SSC ASG, 2016c, 2023), or Vulnerable (Cuvier's gazelle; IUCN SSC ASG, 2016b). Today, most of their ranges are reduced to a few small patches, largely due to human hunting pressure and habitat loss (Newby, 1980; Devillers and Devillers-Terschuren, 2006), and wild populations of addax and dama are estimated to contain fewer than 200 individuals each (IUCN SSC ASG, 2016a, d).

Several range countries have initiated large-scale conservation initiatives for Sahelo-Saharan antelope (SSA). Addax have been reintroduced into fenced sections of multiple protected areas (since 1994; Engel and Brunsing, 1999; Müller and Engel, 2004) and unfenced reserves in Morocco (since 2018; ANEF, 2022) and Tunisia (since 1985; Gordon, 1988; Kacem et al., 1994; Gilbert, 2004; Gilbert et al., 2018), and into the large, unfenced Réserve de Faune du Ouadi Rimé-Ouadi Achim (RFOROA) in Chad since 2020 (Wacher et al., 2022). Cuvier's gazelle were reintroduced into a fenced section of Tergou and Chekhar Reserves and Ifrane National Park in Morocco (since 2018; IUCN, 2018; Hanane and Amhaouch, 2021; ANEF, 2022) and fenced sections of several protected areas in Tunisia (since 1999; Cano and Abáigar, 2002; Abaigar et al., 2005; IUCN, 2018; Moreno et al., 2018, 2020). Population reinforcements or reintroductions of dama gazelle have been conducted in fenced sections of multiple protected areas (since 1992; Wiesner and Müller, 1998; Müller, 2002; Cuzin et al., 2007; IUCN, 2018; Abaígar et al., 2019) and an unfenced reserve (since 2015; IUCN, 2018; Abaígar et al., 2019) in Morocco, fenced areas within the Réserve Spéciale de Faune de Guembeul (RSFG) and Réserve de Faune du Ferlo Nord (RFFN) in Senegal (since 1984; Cano et al., 1993; Akakpo et al., 1995; Jebali, 2005, 2008; Abaígar et al., 2009; Moreno et al., 2012), and fenced sections of multiple protected areas in Tunisia (since 1990; Abaigar et al., 1997; Wiesner and Müller, 1998; Jebali and Zahzah, 2013; Jebali et al., 2019). Oryx have been reintroduced into the RFOROA in Chad (since 2016; Chuven et al., 2018; Mertes et al., 2019), fenced sections of several protected areas in Morocco (since 1995; Engel and Brunsing, 1999, Műller, 2002; Müller and Engel, 2004; Woodfine et al., 2005; Gilbert et al., 2012; ANEF, 2022), a fenced section of RFFN in Senegal (since 1999; Ba and Clark, 2003; Clark, 2002; Gilbert, 2004), and fenced sections of multiple protected areas in Tunisia (since 1985; Wacher, 1986; Gordon, 1991; Gordon and Gill, 1993; Kacem et al., 1994; Molcanova, 2004; Woodfine et al., 2009; Gilbert et al., 2018).

The long-term viability of extant and reintroduced populations of these SSA remains uncertain. Forty-four reintroduced oryx were lost in a 2018 mass mortality event (MME) in Chad, constituting a 53% mortality rate of the adult oryx released that year, greatly exceeding that of previous releases (Chardonnet, 2019; Chardonnet and Nare, 2022; *T. Wacher and K. Mertes, unpublished data*). In combination with nutritional stress and exceptional weather conditions, infectious diseases were a contributing factor: five of nine oryx tested during this event were positive by PCR and/or ELISA for Rift Valley fever (RVF), a

vector-borne viral zoonosis with a broad ruminant host range (Chardonnet, 2019; Chardonnet and Nare, 2022; World Organisation for Animal Health, 2019). Other findings included *Pasteurella multocida*, *Babesia bigemina*, and infestations of hematophagous flies (primarily *Haematobia* spp.) and *Hyalomma* ticks (Chardonnet, 2019; Chardonnet and Nare, 2022). This MME marked the first formal recognition of RVF in Chadian wildlife by the World Organisation for Animal Health (WOAH, formerly OIE; WOAH, 2019). However, serologic evidence of RVF had been previously reported in livestock, humans, and wild ruminants in Chad (Maurice, 1967; Ringot et al., 2004; Abakar et al., 2014), and human outbreaks had been reported in neighboring Sudan (Hassan et al., 2011), indicating the virus may be locally circulating in livestock and/or endemic in these regions.

As was the case for the 2018 oryx MME in Chad, immunologically naïve animals intended for conservation translocations may be susceptible to diseases at the destination site. Spatiotemporal overlap with livestock has been known to precipitate disease-related declines in sympatric wildlife. For example, an epizootic of peste des petits ruminants (PPR) in Mongolia in 2016–2017 killed thousands of critically endangered saiga antelope (*Saiga tatarica mongolica*), as well as smaller numbers of other rare wild ungulates, and likely occurred through exposure to infected livestock (Pruvot et al., 2020). In the USA, outbreaks of pneumonia in bighorn sheep (*Ovis canadensis*)—some resulting in near 100% herd mortality—often follow contact with domestic sheep infected by *Mycoplasma ovipneumoniae* (Besser et al., 2012). Examples of conservation initiatives impeded by infectious diseases include translocated moose acquiring the meningeal worm *Parelaphostrongylus tenuis* carried by indigenous white-tailed deer (*Odocoileus virginianus*) (Anderson, 1972; Aho and Hendrickson, 1989), and squirrelpox outbreaks in red squirrels (*Sciurus vulgaris*) following introduction into areas inhabited by seropositive gray squirrels (*Sciurus carolinensis*) (Carroll et al., 2009). In sub-Saharan Africa, bovine tuberculosis (bTB) and brucellosis (*Brucella abortus*, *B. melitensis*) were likely historically introduced via the movement of domestic cattle, thereafter becoming endemic in some wild ungulate populations (Bengis et al., 2023).

However, transmission pathways are bidirectional between wildlife and domestic livestock. The potential for translocated animals to introduce exotic pathogens to extant conspecifics or livestock is a recognized risk of conservation translocations. Multi-species infectious diseases can be maintained in wildlife populations with occasional spillover into livestock, driving regular sylvatic cycles (Bengis et al., 2023). Wildlife as reservoir or amplification hosts can contribute to pathogen maintenance and re-infection of domestic animals, thus confounding control efforts (Van Vuuren and Penzhorn, 2015; Jori et al., 2021). In sub-Saharan Africa, wildlife can be a significant component of the epidemiology of foot-and-mouth disease (FMD), bTB, trypanosomiasis (including *nagana* in cattle and *sura* in camels), and brucellosis (De Garine-Wichatitsky et al., 2013; Miguel et al., 2013; Weaver et al., 2013; Di Nardo et al., 2015; WOAH, 2021; Bengis et al., 2023).

Anthropogenic factors such as extensive livestock grazing, permeable fencing, habitat fragmentation, and mobile agropastoralism (common across much of Central and North Africa) can increase the probability of contacts between wildlife, livestock and humans and thus facilitate opportunities for spillover (Bengis et al., 2002; Caron et al., 2013; Appolini et al., 2019; Jori et al., 2021; Magiri et al., 2021). Human population growth, rising demand for livestock products, and development programs—such as the drilling of borehole wells—have intensified human use of ever more arid segments of the Sahel and Sahara (De Leeuw and Tothill, 1990). The limited availability of high-quality foraging patches, shaded areas, and water sources across these arid landscapes increases the probability of direct encounters through shared use of scarce resources (VanderWaal et al., 2017; Manso-Silván and Thiacourt, 2019; Titcomb, 2020; Wacher et al., 2022).

Current guidelines for conservation translocations strongly encourage disease risk analyses (IUCN/SSC, 2013; Jakob-Hoff et al., 2014; Hartley and Sainsbury, 2017). Literature reviews can be an important first step to understand disease threats at the destination ecosystem and inform vaccination protocols, screening, and other preventative health measures (D'Alterio et al., 2002; Flach, 2004). In this review, we aimed to compile potential infectious disease risks and knowledge gaps relevant to SSA in immediate danger of extinction and domestic livestock at current and prospective sites for SSA conservation translocations.

Methods

A literature review was carried out during April–December 2020 and retrospectively registered through Inplasy (https://doi.org/10.37766/inplasy2024.10.0039). Ten thematic categories guided keyword selection: "Livestock," "Wild

Ungulates," "Locations," "Infectious Diseases," "Death," "Morbidities," "Outbreaks," "Pathogens," "Epidemiology," and "Diseases–Priority" (S1). Simple and complex search strings comprising three categories were used (S1). Results were retrieved from four databases: Web of Science Core Collection, ProQuest, PubMed, and Zoological Record. Each database was queried 27 times, for a total of 108 searches targeting reports from 2000 to 2020.

An initial geographically broad search was carried out for English-language articles pertaining to the historic ranges of all SSA. A second series of searches refined "Locations" to only Chad, Morocco, Senegal, and Tunisia, to focus on regions with active or prospective SSA conservation translocation initiatives. An additional search of the four databases using the refined geographic range, time period, and search strings was performed for French-language articles.

Citations were managed using Zotero (Corporation for Digital Scholarship, 2021). During the first review round, one or two independent reviewers (LE, JHY for English articles, ID for French) assessed each record's metadata for relevance (reports of infectious diseases in wild and domestic ruminants and camelids), with any disputes resolved by a third party (DZ). For citations of uncertain relevance based on metadata alone, the full article was assessed (Fig. 1).

During the second review round, one independent reviewer (LE, JHY for English articles, ID for French) read full articles to determine relevance. Inaccessible papers not available for full review were excluded. Records not meeting inclusion criteria for geographic range (Chad, Morocco, Senegal, Tunisia), topic, language, or host species (wild or domestic ruminants and camelids) were excluded. Papers discussing exclusively human cases, experimental infections, predictive modeling, vector or parasite ecology, antimicrobial resistance, or food-borne pathogens or mastitis (unless systemic) were excluded. Reviews with no new data and studies with duplicate findings were also excluded (Fig. 1).

An additional search was conducted in the WOAH World Animal Health Information System (WAHIS) database, which has collated global data on WOAH-listed animal diseases since 2005 (wahis.woah.org). For each country of interest, records through Dec 2020 pertaining to SSA and domestic livestock were included. Each record detailed an initial "disease event," with weekly outbreak updates until resolution; each disease event was considered a discrete report. Finally, infectious agents were classified into broad functional and taxonomic categories: "viruses," "protozoa," "bacteria," "fungal/other," "ectoparasites," and "endoparasites" (Tables 1, 2, 3, 4, 5, 6).

Results

We retrieved 26,047 papers (25,632 in English and 415 in French) through our peer-reviewed literature search. After deduplication, 6071 results (5911 in English and 160 in French) remained. After initial screening, 328 relevant English-language publications and 32 relevant French articles remained. Following the second review round, 202 records (182 English, 20 French) were deemed relevant (Fig. 1, S2).

The majority of results were from Tunisia (52.3%, $n = 106$), followed by Morocco (21.3%, $n = 43$), Senegal (17.3%, $n = 35$) and Chad (9.4%, $n = 19$, Figures 2, 3a). Publication dates ranged from 2002 to 2020, with a general increase across all countries beginning around 2013, which roughly coincides with the increased usage of molecular-informed methods, such as PCR and phylogenetic analysis (Fig. 3b). Most studies (120, 59.4%) used molecular techniques, followed by serology (84, 41.6%). Other reported diagnostics included bacterial culture, viral isolation, microscopy with morphological identification, post-mortem examination, histopathology, ultrasonography, or specialized tests (e.g., single intradermal comparative cervical tuberculin test). Cattle (106, 52.5%) and sheep (100, 49.5%) were the most-represented hosts by publication, followed by goats (69, 34.2%) and camels (35, 17.3%).

Only six publications (3.0%) reported infectious agents in wildlife. Three articles specifically mentioned species of interest for this review: addax, oryx, or slender-horned gazelle (Tables 1, 2, 6). Other wildlife host species reported were dorcas gazelle (*Gazella dorcas*), Barbary red deer (*Cervus elaphus barbarus*), Barbary sheep (*Ammotragus lervia*), and western Derby eland (*Taurotragus derbianus*) living in protected areas. Reported infectious agents primarily included gastrointestinal helminths and protozoal parasites (Tables 1, 2, 6). Notably, however, one paper confirmed eight cases (including three mortalities) of RVF in dorcas gazelle from the Réserve Spéciale de Faune de Gueumbeul (RSFG) during a widespread epizootic and epidemic in Senegal during 2013–2014 (WOAH, 2013; Sow et al., 2016).

Bacterial pathogens were discussed in 63 studies (31.2%), protozoal pathogens in 51 (25.2%), fungal agents

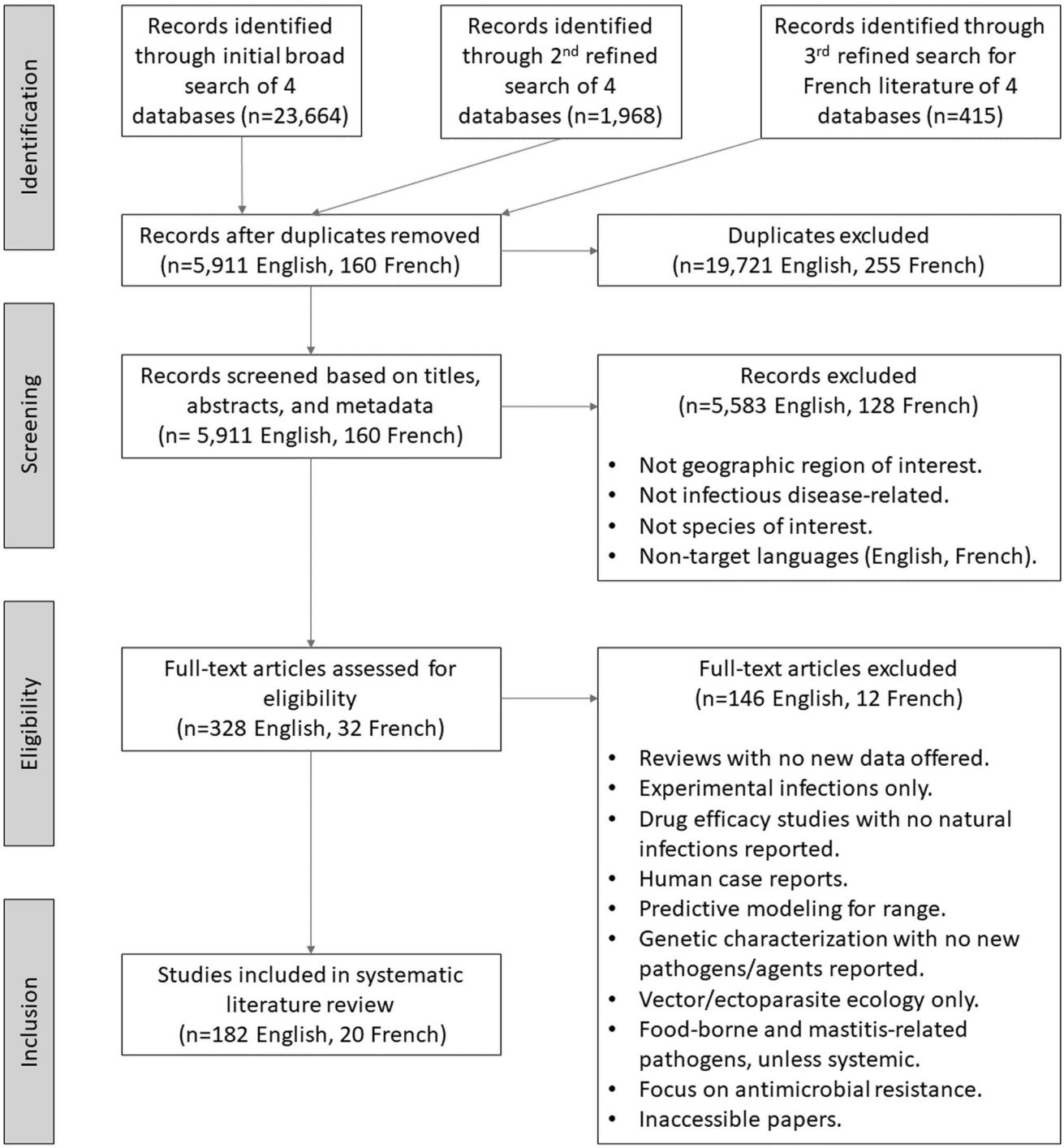


Figure 1. Overview of systematic literature review methodology and results, presented in PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) format.

in two (0.99%), and viruses in 54 (26.7%) (Tables 1, 2, 3, 4, 5 6). Twenty-eight publications (13.9%) concerned ectoparasites, while 29 studies (14.4%) featured endoparasites. Nearly one-third of all relevant publications (64, 31.7%) discussed infectious agents for which arthropod vectors play a large role in transmission.

The WOAH-WAHIS database returned 38 relevant results based on host species and country of interest. Two reports were retrieved from Chad, 11 from Morocco, five from Senegal, and 20 from Tunisia. Multiple outbreaks were reported for several key livestock diseases: PPR and bluetongue in goats and sheep in Morocco and Tunisia;

Table 1. Viral Pathogens and Infectious Agents of Domestic Livestock and Wild Ungulates.

Order	Family	Infectious Agent	Host Species Reported	Regions Reported	References
Amarillovirales	Flaviviridae	Border disease virus (BDV) (pestivirus D)	G, Sh	M	79, 153
		Bovine viral diarrhea virus (BVDV)	Ct	M, S, T	108, 118, 194
		Tick-borne encephalitis virus (TBEV)	Sh	T	105
		West Nile virus (WNV)	C, Ct, G, Sh	M, S, T	60, 93, 195
Articulavirales	Orthomyxoviridae	Influenza D virus	Ct	M	166, 171
Bunyavirales	Nairoviridae	Crimean-Congo hemorrhagic fever virus (CCHFV)	C, Ct, G, Sh	S, T	46, 121
		Dugbe virus (DUGV)	C	Ch	161
	Phenuiviridae	Rift Valley fever virus (RVFV)[1]	C, Ct, G, S, W[1]	Ch, S, T	1, 19, 53, 54, 55, 72, 93, 156, 191, WAHIS
Chitovirales	Poxviridae	Camelpox virus (CMPV)	C	M	195
		Sheeppox virus (SPV)	Sh	M, T	110, 201, 202
Herpesvirales	Herpesviridae	Bovine alphaherpesvirus-1 (BHV-1)	Ct	M, S	108, 118
		Bovine gammaherpesvirus-4 (BHV-4)	Ct	M	118
Mononegavirales	Paramyxoviridae	Small ruminant morbillivirus (peste-des-petits-ruminants virus)	G, Sh	Ch, M, S, T	19, 34, 68, 78, 109, 120, 135, 170, 185, WAHIS
Nidovirales	Coronaviridae	Dromedary camel coronavirus HKU23	C	M	188
		Middle East respiratory syndrome coronavirus (MERS-CoV)	C, Sh	M, S, T	98, 99, 132, 155
Picornavirales	Caliciviridae	Nebovirus sp.	Ct	T	95
		Norovirus sp. (Group III)	Ct	T	95
	Picornaviridae	Foot-and-mouth disease virus (FMDV)	C, Ct, G, Sh	Ch, M, T	3, 145, WAHIS
Reovirales	Reoviridae	African horse sickness virus (AHSV)	C	M	195
		Bluetongue virus (BTV)[2]	C, Ct, G, Sh	M, S, T	17, 70, 82, 93, 117, 186, 195, WAHIS
		Epizootic hemorrhagic disease virus (EHDV)	C, Ct	M, T	33, 93, 125, WAHIS
		Rotavirus sp.[3]	Ct, G	M, T	8, 76, 80, 94, 116
Zurhausenvirales	Papillomaviridae	Bovine papillomavirus-1 (BPV-1)	Ct	M	150

C camels, *Ct* cattle, *G* goats, *Sh* sheep, *W* wildlife, *Ch* Chad, *M* Morocco, *S* Senegal, *T* Tunisia.
[1]Reported in Dorcas gazelles (*Gazella dorcas*), scimitar-horned oryx (*Oryx dammah*), roan antelope (*Hippotragus equinus*).
[2]Multiple genotypes reported (BTV-1,-2,-3,-4,-6,-8,-14,-16).
[3]Includes Group A rotaviruses.

FMD in cattle and small ruminants in Morocco and Tunisia; and contagious bovine pleuropneumonia (CBPP; caused by *Mycoplasma mycoides*) in cattle in Senegal (Table 7). Two pathogens reported to WAHIS were not captured in our review of peer-reviewed literature: *Ehrlichia ruminantium* and the prion protein isoform PrPSc, causative agent of scrapie. Notably, along with the previously mentioned RVF outbreaks in Chad and Senegal, the database reported an additional RVF episode in Senegal in 2020: a roan antelope from a private reserve died from RVF after a period of heavy rainfall. During the same disease event, two RVF cases (including one mortality) in dorcas

Table 2. Protozoal Pathogens and Infectious Agents of Domestic Livestock and Wild Ungulates.

Phylum	Class	Order	Family	Infectious Agent	Host Species Reported	Regions Reported	References
Apicomplexa	Aconoidasida	Piroplasmida	Babesiidae	*Babesia bigemina*	Ct	M, T	73, 92, 127, 128
				Babesia bovis	Ct	M, T	92, 127, 128
				Babesia ovis	G, Sh	T	157, 160
				Babesia sp.	C, Sh	T	165, 183
			Theileriidae	*Theileria annulata*	Ct	M, T	52, 59, 73, 74, 84, 92, 128, 131, 172, 197
				Theileria buffeli	Ct	T	128
				Theileria lestoquardi	Sh	T	159
				Theileria ovis	G, Sh	T	130, 157, 160
				Theileria sp.	Sh	T	165
	Conoidasida	Eucoccidiorida	Cryptosporidiidae	*Cryptosporidium bovis*	Sh	T	190
				Cryptosporidium parvum	Ct	T	147, 189
			Eimeriidae	*Eimeria derbiani*[1]	W[2]	S	119
				Eimeria sp.	Sh	M	146
			Sarcocystidae	*Neospora caninum*	Ct, G, Sh	M, S, T	13, 14, 16, 58, 97, 118
				Sarcocystis capracanis	G	T	10
				Sarcocystis cruzi	Ct	T	9
				Sarcocystis hominis	Ct	T	9
				Sarcocystis tenella	Sh	T	10
				Toxoplasma gondii	Ct, G, Sh	M, S, T	11, 15, 27, 43, 49, 58, 61, 77, 85, 107, 153, 163, 164, 174
Euglenozoa	Kinetoplastea	Trypanosomatida	Trypanosomatidae	*Trypanosoma brucei*[3]	Ct	Ch, S	138, 176
				Trypanosoma congolense	Ct	S	149, 176
				Trypanosoma evansi	C, Ct	Ch, M, T	18, 63, 172
				Trypanosoma vivax	Ct, G, Sh	Ch, S	64, 81, 138, 149, 176
				Trypanosoma sp.	C	M, T	148, 183

C camels, *Ct* cattle, *G* goats, *Sh* sheep, *W* wildlife, *Ch* Chad, *M* Morocco, *S* Senegal, *T* Tunisia.

[1]Proposed new species.

[2]Reported in semi-captive western Derby eland (*Taurotragus derbianus derbianus*).

[3]Includes *T. brucei* sensu lato.

Table 3. Bacterial Pathogens and Infectious Agents of Domestic Livestock and Wild Ungulates.

Phylum	Class	Order	Family	Infectious Agent	Host Species Reported	Regions Reported	References
Actinobacteria	Actinomycetia	Corynebacteriales/ Mycobacteriales	Corynebacteriaceae	*Corynebacterium pseudotuberculosis*	Sh	T	42
			Mycobacteriaceae	*Mycobacterium avium* subsp. *paratuberculosis*	Sh	T	106
				Mycobacterium avium complex[1]	Ct	Ch	67
				Mycobacterium bovis	Ct	Ch, M, T	35, 65, 66, 69, 114, 142, 198, 199
				Mycobacterium caprae	Ct	T	114
				Mycobacterium simiae	Ct	Ch	67
				Mycolicibacter nonchromogenicus[2]	Ct	Ch	67
				Mycolicibacterium fortuitum[3]	Ct	Ch	67
			Nocardiaceae	*Nocardia sp.*	Sh	T	42
Chlamydiota	Chlamydia	Chlamydiales	Chlamydiaceae	*Chlamydia abortus*	Ct, G, Sh	M, T	22, 26, 43, 62, 151, 152, WAHIS
				Chlamydia pecorum	Ct, Sh	T	26
				Chlamydia sp.	G, Sh	T	153
		Parachlamydiales	Parachlamydiaceae	*Parachlamydia acanthamoebae*	Ct, Sh	T	22, 26
			Waddliaceae	*Waddlia chondrophila*	Ct	T	21, 22
Firmicutes	Bacilli	Bacillales	Bacillaceae	*Bacillus anthracis*	Ct	Ch	2, 192
			Listeriaceae	*Listeria monocytogenes*	Ct, Sh	T	22, 23
			Staphylococcaceae	*Staphylococcus aureus*	Ct, G	T	86
				Staphylococcus aureus subsp. *anaerobius*	Sh	T	42
		Lactobacillales	Streptococcaceae	*Streptococcus sp.*	Sh	T	42
Mycoplasmatota	Mollicutes	Mycoplasmatales	Mycoplasmataceae	*Mycoplasma ovis*	Sh	T	158
				Mycoplasma mycoides	Ct	S	123, WAHIS
Proteobacteria	Alphaproteobacteria	Hyphomicrobiales	Bartonellaceae	*Bartonella bovis*	Ct	S	57
				Bartonella henselae	C	T	178
				Candidatus *Bartonella davousti*	Ct	S	57
			Brucellaceae	*Brucella abortus*	Ct, G, Sh	M, S, T	22, 108, 199
				Brucella melitensis	Ct	S	108
				Brucella sp.	C, Ct, G, Sh	Ch, M, S, T	1, 25, 43, 104, 118, 137, 153, 175

Table 3. continued

Phylum	Class	Order	Family	Infectious Agent	Host Species Reported	Regions Reported	References
		Rickettsiales	Anaplasmataceae	*Anaplasma bovis*	Ct, G, Sh	T	28, 30, 40
				Anaplasma centrale	Ct	S, T	28, 56
				Anaplasma marginale	Ct, G, Sh	M, S, T	4, 5, 28, 56, 92, 126
				Anaplasma ovis	C, G, Sh	M, S, T	5, 29, 30, 37, 56, 180
				Anaplasma phagocytophilum	C, Ct	T	41, 126
				Anaplasma phagocytophilum-like strains	Ct, G, Sh	T	37, 39
				Anaplasma platys	Ct, G, Sh	T	38
				Anaplasma platys-like strains	C, Ct, G, Sh	S, T	32, 56, 181
				Anaplasma sp. 'Omatjenne'	C, Ct, Sh	M, T	183, 193
				Candidatus *Anaplasma africae*	Ct, G, Sh	S	56
				Candidatus *Anaplasma camelii*	C	M	6
				Ehrlichia ruminantium	C	Ch	WAHIS
			Rickettsiaceae	*Rickettsia aeschlimannii*	C	T	179
				Rickettsia africae	C	T	179
				Rickettsia helvetica	C	T	179
				Rickettsia massiliae	C	T	179
				Rickettsia monacensis	C	T	179
	Betaproteobacteria	Burkholderiales	Alcaligenaceae	*Bordetella bronchiseptica*	G, Sh	S	143
	Gammaproteobacteria	Enterobacterales	Enterobacteriaceae	*Escherichia coli*	C	T	45
				Klebsiella oxytoca	Sh	T	42
				Salmonella enterica serovar Abortus	G, Sh	T	153
				Salmonella enterica serovar Typhimurium	Ct	T	22
		Legionellales	Coxiellaceae	*Coxiella burnetii*	C, Ct, G, Sh	Ch, M, T	1, 24, 43, 153, 175, 184, WAHIS

Table 3. continued

Phylum	Class	Order	Family	Infectious Agent	Host Species Reported	Regions Reported	References
Spirochaetota	Spirochaetia	Leptospirales	Leptospiraceae	*Leptospira sp.*[4]	Ct, G, Sh	M, S, T	43, 44, 102, 103, 118, 162
		Spirochaetales	Borreliaceae	*Borrelia burgdorferi*	C, Ct, G, Sh	T	36

C camels, *Cr* cattle, *G* goats, *Sh* sheep, *W* wildlife, *Ch* Chad, *M* Morocco, *S* Senegal, *T* Tunisia.
[1]Includes one strain of *Mycobacterium avium-intracellulaire*.
[2]Previously *Mycobacterium nonchromogenicum*.
[3]Previously *Mycobacterium fortuitum*.
[4]Reported species include *L. interrogans* and *L. wolffii*. Reported serovars include Australis, Autumnalis, Ballum, Bim, Bratislava, Canicola, Copenhageni, Grippotyphosa, Hardjo, Hebdomadis, Icterohaemorrhagiae, Munchen, Mozdok, Panama, Pomona, Pyrogenes, Saxkoebing, Sejroe, and Tarassovi.

gazelle (*Gazella dorcas*) were also confirmed in RSFG (WOAH, 2020).

In total, 159 distinct infectious agents were reported in the literature and WOAH-WAHIS database (Tables 1, 2, 3, 4, 5, 6), consisting of bacteria (45, 28.3%), nematodes and platyhelminthes (36, 22.6%), ectoparasites (33, 20.8%), viruses (23, 14.5%), protozoa (20, 12.6%), one fungus (1, 0.63%), and one prion (1, 0.63%). Excluding ectoparasites, many of which are considered anthropophilic to some extent, 61 of the 126 remaining distinct infectious agents (48.4%) were potentially zoonotic, including both established zoonoses and agents for which human infections have been reported but may be rare or subclinical. Thirty distinct infectious agents were considered WOAH-listed diseases (WOAH, 2021) (Tables 8, 9).

The infectious agent and/or parasite taxa most frequently recorded in peer-reviewed publications across our target countries were *Anaplasma spp.* (n = 19 studies), *Theileria spp.* (n = 15), *Toxoplasma gondii* (n = 14), *Brucella spp.* (n = 11), *Mycobacterium spp.* (n = 10), and *Trypanosoma spp.* (n = 10). The most frequently reported ectoparasites were *Hyalomma spp.* (n = 16 studies) and *Rhipicephalus spp.* (n = 12). When considered in totality, the WOAH-WAHIS reports reflected a more complete timeline of discrete disease events and increased the number of records of bluetongue (n = 19), PPR (n = 16), and RVF (n = 12). Other commonly reported agents included *Chlamydia spp.* (n = 9 records), *Babesia spp.* (n = 8), *Echinococcus spp.* (n = 8), *Coxiella burnetii* (n = 7), *Leptospira spp.* (n = 6), and *Neospora caninum* (n = 6). The most frequently reported agents and pathogens by country and functional/taxonomic group are depicted in Figure 4.

Discussion

The aim of our literature review was to identify infectious agents of concern for Sahelo-Saharan antelope (SSA) considered at immediate risk of extinction in range states with active or planned conservation translocation programs. This review highlighted a number of infectious agents with broad potential host ranges and of varying veterinary, public health, and production importance, including FMD virus, bluetongue virus (BTV), RVF, PPR, *Anaplasma spp.*, *Brucella spp.*, *Leptospira spp.*, *Coxiella burnetii*, *Mycoplasma spp.*, *Mycobacterium spp.*, *Eimeria spp.*, *Borrelia burgdorferi*, *Fasciola spp.*, *Trypanosoma spp.*, *Haemonchus spp.*, and *Schistosoma spp.*

Table 4. Fungal and Other Pathogens and Infectious Agents of Domestic Livestock and Wild Ungulates.

Order	Family	Infectious Agent	Host Species Reported	Regions Reported	References
Onygenales	Arthrodermataceae	*Trichophyton sarkisovii*	C	M	75
		Unknown "ringworm" species[1]	C	M	71
–-	–-	PrPSc (scrapie prion)	Sh	T	WAHIS

C camels, *Ct* cattle, *G* goats, *Sh* sheep, *W* wildlife, *Ch* Chad, *M* Morocco, *S* Senegal, *T* Tunisia.
[1]Species not identified or named in reference.

Several of the pathogens highlighted in this review have the capacity to cause morbidity and mortality in wild ruminants. The abortifacient *C. burnetii*, which causes Q fever, has a wide host range but is primarily known to cause reproductive losses in domestic ruminants. Considered endemic in Africa (Salifu et al., 2019), *C. burnetii* has been implicated in reproductive losses in wild ruminants including water buffalo (*Bubalus bubalis*), waterbuck (*Kobus ellipsiprymnus*), roan antelope (*Hippotragus niger*), and several gazelle, including captive dama (González-Barrio and Ruiz-Fons, 2018). Antibodies to *C. burnetii* have also been reported in Arabian oryx (*Oryx leucoryx*), although the clinical significance was unclear (Greth et al., 1992).

Similarly to Q fever, PPR is most commonly associated with disease in small ruminants, resulting in morbidity and mortality rates of up to 100% and 90%, respectively (Kumar et al., 2014). Some wild species can be similarly affected; PPR has been associated with clinical disease in free-ranging goitered gazelle (*Gazella subgutturosa*), ibex (*Capra ibex sibirica*), and wild goats and sheep (*Ovis orientalis*, *O. vignei*) (Fine et al., 2020). A PPR outbreak in critically endangered saiga antelope (*Saiga tatarica mongolica*) in Mongolia caused a population decline of approximately 80% (Pruvot et al., 2020). However, the clinical effects in African wildlife appear less pronounced: while PPR seropositivity has been reported widely in numerous wild African ruminant species, the only reports to date of clinical disease in African wildlife to date were in captive and potentially free-ranging dorcas gazelles (*Gazella dorcas*) in Sudan, although the latter has been questioned (Asil et al., 2019; Fernandez Aguilar et al., 2020; Jones et al., 2021). The differential susceptibility to PPR observed among wild artiodactyls is not well understood (Fine et al., 2020).

Our review yielded relatively few relevant reports in wild ungulates, but did reveal additional mortalities due to RVF in dorcas gazelle in Réserve Spéciale de Faune de Guembeul (RSFG) and a roan antelope (*Hippotragus equinus*) in a private reserve in Senegal, which were suspected to be associated with concurrent disease events in livestock (WOAH, 2013, 2020; Sow et al., 2016). Given the geographic distribution of cases, Sow et al. (2016) proposed that mass transportation and growing animal markets associated with a major religious event contributed to this outbreak (WOAH, 2013). Because RVF is mosquito-borne, outbreaks occur sporadically and depend on environmental factors like water availability. In East Africa, RVF outbreaks have been linked to flooding and El Niño-Southern Oscillation cycles. During quiescent, interepizootic periods, the RVF virus is thought to circulate at low levels in livestock and wildlife hosts (Paweska, 2015; Rostal et al., 2017). In semi-arid landscapes like the Sahel, the factors driving RVF emergence and endemicity are less well-studied: environmental and climatic factors including surface water persistence and rainfall frequency and intensity may favor vector population growth. Inter-site variability in rainfall drives herd movements, and the convergence of livestock and wildlife hosts on vegetation patches when vectors are abundant may drive emergence (Favier et al., 2006; Cecilia et al., 2020; Lancelot et al., 2019). Thus, the co-occurrence of domestic and wild ruminants may contribute to amplification during epidemics, or maintenance between outbreaks. The clinical course of RVF in wildlife can range from asymptomatic infections to abortion to death, suggesting variability in host species susceptibility, or that other factors may drive pathogenicity and virulence (Paweska, 2015; Rostal et al., 2017).

Other infectious agents reported in wild ruminants in target countries included *Eimeria derbiani*, *Echinococcus granulosus*, and multiple strongylids. In 2021, Saidi et al. investigated parasites affecting the semi-captive populations of addax, oryx, and dorcas gazelle in Souss Massa National Park, Morocco. Their findings revealed a high prevalence of Coccidians (specifically *Eimeria sp.*), as well as pulmonary protostrongylids, including *Muellerius capillaris* and *Neostrongylus linearis*, along with gastrointestinal nematodes such as *Nematodirus spathiger* and

Table 5. Ectoparasites of Domestic Livestock and Wild Ungulates.

Order	Family	Ectoparasite	Host Species Reported	Regions Reported	References
Diptera	Ceratopogonidae	*Culicoides imicola*	Ct, G, Sh	T	187
		Culicoides jumineri	Ct, G, Sh	T	187
		Culicoides newsteadi	Ct, G, Sh	T	187
	Culicidae	*Anopheles gambiae*	Ct, Sh	S	144
	Psychodidae	*Phlebotomus chabaudi*	C, Ct, G, Sh	T	96
		Phlebotomus longicuspis	C, Ct, G, Sh	T	96, 154
		Phlebotomus papatasi	C, Ct, G, Sh	T	96
		Phlebotomus perfiliewi	Ct, G, Sh	T	154
		Phlebotomus perniciosus	Ct, G, Sh	T	154
		Phlebotomus sergenti	C, Ct, G, Sh	T	96
		Sergentomyia minuta	C, Ct, G, Sh	T	96
	Sarcophagidae	*Wohlfahrtia magnifica*	G, Sh	M	89
Ixodida	Ixodidae	*Haemaphysalis punctata*	Ct	T	129
		Hyalomma detritum	Ct	M, T	101, 129, 173
		Hyalomma dromedarii	C, Ct, Sh	M, T	101, 157, 158, 178, 179, 180, 181, 182, 183
		Hyalomma excavatum	C, Ct, Sh	T	158, 159, 173, 178, 180, 181, 182, 183
		Hyalomma impeltatum	C	T	178, 179, 180, 181, 182, 183
		Hyalomma marginatum	Ct	M, T	101, 173
		Hyalomma rufipes	C, Ct, G, Sh	Ch, S	56, 121, 124, 136
		Hyalomma truncatum	Ct, G, Sh	S	124
		Ixodes ricinus	C	T	129
		Rhipicephalus annulatus	Ct, G, Sh	T	31, 129
		Rhipicephalus bursa	Ct, G, Sh	T	31, 129, 130
		Rhipicephalus camicasi	Sh	T	158, 159
		Rhipicephalus evertsi	Ct, G, Sh	S	56, 121, 124
		Rhipicephalus guilhoni	Ct, G, Sh	S	124
		Rhipicephalus sanguineus	G, Sh	T	31, 105, 158, 159
		Rhipicephalus turanicus	Ct, G, Sh	T	31, 129, 130, 157, 158, 159, 160, 183
Psocodea	Haematopinidae	*Haematopinus eurysternus*	Ct	T	83
	Linognathidae	*Linognathus vituli*	Ct	T	83

Table 5. continued

Order	Family	Ectoparasite	Host Species Reported	Regions Reported	References
	Trichodectidae	*Bovicola bovis*	Ct	T	83
Sarcoptiformes	Sarcoptidae	*Sarcoptes scabiei*	C	M	71
Siphonoptera	Pulicidae	*Ctenocephalides felis*	G, Sh	T	200

C camels, *Ct* cattle, *G* goats, *Sh* sheep, *W* wildlife, *Ch* Chad, *M* Morocco, *S* Senegal, *T* Tunisia.

Camelostrongylus mentulatus. Despite the observed prevalence, the overall infestation rates were relatively low, indicating that these parasites likely do not pose a significant threat to survival for these species (Saidi, 2021). As single agents, gastrointestinal parasites generally do not cause notable disease in ruminants, but can in some circumstances result in general ill thrift and increased susceptibility to co-infections. High parasite burdens or chronic infections can cause more severe pathology. For example, ruminants are usually intermediate hosts for *E. granulosus*, remaining asymptomatic until cysts become large enough to invade and cause damage to adjacent tissues (Spickler, 2020).

While not frequently reported in the peer-reviewed literature for our countries of interest, WOAH-listed foot-and-mouth disease (FMD) and paratuberculosis—both diseases of global economic importance—have been responsible for outbreaks in captive oryx in the United Arab Emirates (UAE) and Italy, respectively (Lignereux et al., 2020; Pigoli et al., 2020). Caused by a highly infectious aphthovirus, FMD is a transboundary disease of artiodactyls, with a wide range of clinical presentations in wildlife. The FMD outbreak in scimitar-horned oryx in the UAE was characterized by rapid transmission and high seroprevalence (peaking at 78%); clinical signs in oryx were largely consistent with those in livestock and included mucosal erosions and cutaneous lesions at the horn base and coronary bands (Lignereux et al., 2020). Although mortality was not observed as a direct result of the virus, FMD could have serious repercussions if introduced into naïve populations, due to its high transmissibility and effects on body condition. FMD has been found circulating in African buffalo and other wild ruminants in Chad, underscoring the risk of exposure from both livestock and wildlife (Di Nardo et al., 2015). Paratuberculosis, caused by *Mycobacterium avium* subspecies *paratuberculosis*, typically takes a more insidious approach, with the onset of clinical signs appearing years after initial infection. In livestock, most animals either clear the pathogen or become asymptomatic carriers, but those that develop clinical signs exhibit persistent diarrhea and progressive emaciation, and slowly succumb to the disease. Pathogenesis in wild ruminants is not as well documented, but infected oryx in a captive collection in Italy similarly demonstrated enteritis and emaciation, resulting in the death or euthanasia of 6/10 animals (Pigoli et al., 2020).

Out of 28 publications investigating ectoparasites, 14 (50%) examined vector-borne pathogens found in ectoparasites themselves; primarily ixodid ticks, but also fleas and phlebotomine flies. Seven studies investigated pathogens in both collected vectors and direct samples from livestock. Of these, most studies (6/7, 85.7%) found a lower prevalence of pathogens in ectoparasite vectors versus livestock. The sole exception to this pattern reported *Coxiella burnetii* in *Hyalomma* ticks, but not in camels by PCR (Selmi et al., 2019). Not all infectious agents identified in ectoparasites are known to be pathogenic in ruminants; however, relevant pathogens included *Anaplasma ovis*, *Babesia bigemina*, and *B. motasi*. Our review suggests that livestock may be occasional hosts for ticks carrying notable zoonotic pathogens, including *Leishmania sp.* and the spotted fever group rickettsioses: *Rickettsia aeschlimannii, R. africae*, *R. conorii conorii*, *R. massiliae*, and *R. monacensis*. A 2014–2020 Moroccan study published after our review period identified several tick species in a semi-captive addax population within a private reserve, including *H. marginatum, H. lusitanicum, Rhipicephalus sanguineus* sensu lato, and *Rhipicephalus pusillus* (Rodríguez et al., 2022). Addax were found to have a low seroprevalence of *Brucella spp.* (0–6%), in contrast to higher rates of *C. burnetti* (36–70%) and bluetongue virus (BTV) (66–92%), possibly reflecting a role of ticks in transmitting common livestock-associated infectious agents. Notably, Crimean-Congo Hemorrhagic Fever Virus (CCHFV) was

Table 6. Endoparasites of Domestic Livestock and Wild Ungulates.

Phylum	Class	Order	Family	Endoparasite	Host Species Reported	Regions Reported	References
Nematoda	Chromadorea	Strongylida	Chabertiidae	*Chabertia ovina*	W[1]	T	168
			Dictyocaulidae	*Dictyocaulus filaria*	Sh	M	146
			Haemonchidae	*Camelostrongylus mentulatus*	W[1]	T	168
				Haemonchus contortus	Ct, G, Sh	M, T	7, 50, 165
				Haemonchus placei	Ct, G, Sh	T	7
				Marshallagia marshalli	Sh, W[1,2]	T	165, 168
				Ostertagia lyrata	Sh	T	165
				Ostertagia occidentalis	Sh	T	165
				Ostertagia ostertagi	Sh	T	165
				Teladorsagia circumcincta	Sh	T	165
				Teladorsagia trifurcata	Sh	T	165
			Molineidae	*Nematodirus helvetianus*	G, Sh, W[1,2,3,4,5]	T	168
				Nematodirus spathiger	G, Sh, W[1,2,5,6]	T	168
				Nematodirus sp.	Sh	M	146
			Protostrongylidae	*Elaphostrongylus cervi*	W[3]	T	167
				Muellerius capillaris	Sh, W[1,2,5]	M	146, 169
				Neostrongylus linearis	W[1,2,5]	M	169
				Varestrongylus sagittatus	W[3]	T	167
			Trichostrongylidae	*Trichostrongylus axei*	Sh	T	165
				Trichostrongylus vitrinus	Sh	T	165
		Rhabditida	Setariidae	*Setaria labiatopapillosa*	Ct	M	134
Platyhelminthes	Cestoda	Cyclophyllidea	Anoplocephalidae	*Avitellina centripunctata*	Ct, Sh	S	139, 140
				Moniezia expansa	G, Sh	S	139
				Stilesia globipunctata	Sh	S	139
				Thysaniezia connochaeti	Ct	S	141
				Thysaniezia ovilla	G, Sh	S	139, 141
			Taeniidae	*Echinococcus canadensis*	C	T	47
				Echinococcus granulosus	C, Ct, G, Sh, W[1]	M, T	20, 47, 48, 51, 111, 112, 113, 133
				Taenia hydatigena	G, Sh	T	100
				Taenia multiceps	Sh	T	12
	Trematoda	Plagiorchiida	Dicrocoeliidae	*Dicrocoelium hospes*	Ct	S	122
			Gastrothylacidae	*Carmyerius marchandi*	Ct	S	177

Table 6. continued

Phylum	Class	Order	Family	Endoparasite	Host Species Reported	Regions Reported	References
			Fasciolidae	*Fasciola gigantica*	Ct	Ch	87, 88
				Fasciola hepatica	Ct, G, Sh	T	90, 91
			Paramphistomidae	*Paramphistomum microbothrium*	Ct	S	177
		Strigeidida	Schistosomatidae	*Schistosoma bovis*	Ct, G, Sh	Ch, S	88, 115, 196
				Schistosoma bovis x *S. curassoni* hybrid	Ct, G, Sh	S	196
				Schistosoma curassoni	Ct, G, Sh	S	115, 196

C camels, *Ct* cattle, *G* goats, *Sh* sheep, *W* wildlife, *Ch* Chad, *M* Morocco, *S* Senegal, *T* Tunisia.
[1]Reported in addax (*Addax nasomaculatus*).
[2]Reported in dorcas gazelle (*Gazella dorcas*).
[3]Reported in Barbary red deer (*Cervus elaphus barbarus*).
[4]Reported in Barbary sheep (*Ammotragus lervia*).
[5]Reported in scimitar-horned oryx (*Oryx dammah*).
[6]Reported in slender-horned gazelle (*Gazella leptoceros*).

detected in 100% of four tick pools (each composed of ten ticks) tested from 2016–2017. Additionally, the presence of various pathogens was detected in adult *H. marginatum*, including *Babesia ovis* by reverse-line blot (RLB) (75%), *Anaplasma spp*. (75%), spotted fever group *Rickettsia spp*. (50%), and *Theileria velifera* and *T. taurotragi* by RLB (25%).

Several of the livestock diseases detected in our review are also major zoonoses of public health interest, with the potential to cause significant health consequences and disease burden in human hosts. These include RVF virus, *Coxiella burnetii*, *Brucella sp*., *Echinococcus sp*., and *Leptospira sp*. Humans are highly susceptible to RVF, which can cause mild symptoms to severe hemorrhagic disease and death (Kahlon et al., 2010). During outbreaks, occupational exposure through contact with sick animals and their products are significant risk factors for acute infection and severe disease (Mohamed et al., 2007; Anyangu et al., 2010). Human infection with *Coxiella burnetii* is also associated with livestock; major sources of transmission are direct contact with secretions and affected tissues, and inhalation of aerosols (Salifu et al., 2019). Brucellosis, caused by multiple species of bacteria belonging to the genus *Brucella,* is another disease where human transmission generally occurs from occupational exposure and consumption of raw milk and milk products (Godfroid et al., 2013; Ducrotoy et al., 2015). While the disease can be debilitating in humans, a lack of pathognomonic symptoms can make it difficult to distinguish from other conditions, leading to underreporting and difficulty determining true epidemiology of the disease (Ducrotoy et al., 2015). Both *Brucella abortus* and *B. melitensis* have been reported in wild ungulates in Africa and are known to be zoonotic (Godfroid et al., 2013; Ducrotoy et al., 2015; Simpson et al., 2021). Tapeworms of the genus *Echinococcus* are globally distributed and constitute a major public health concern in endemic regions. In intermediate hosts, the larval stage forms cysts within the liver, lungs, and other organs, causing structural damage as they grow (Spickler, 2020). Transmission to humans typically occurs via accidental ingestion of eggs shed by carnivore (usually canine) definitive hosts, due to poor food safety habits or direct contact with infected animals. Because livestock serve as intermediate hosts, they play an important role in perpetuating the parasite life cycle and overall risk to human health (Spickler, 2020). Similarly, both domestic and wild animals can serve as reservoirs for the bacterial disease leptospirosis and shed spirochetes environmentally. Hu-

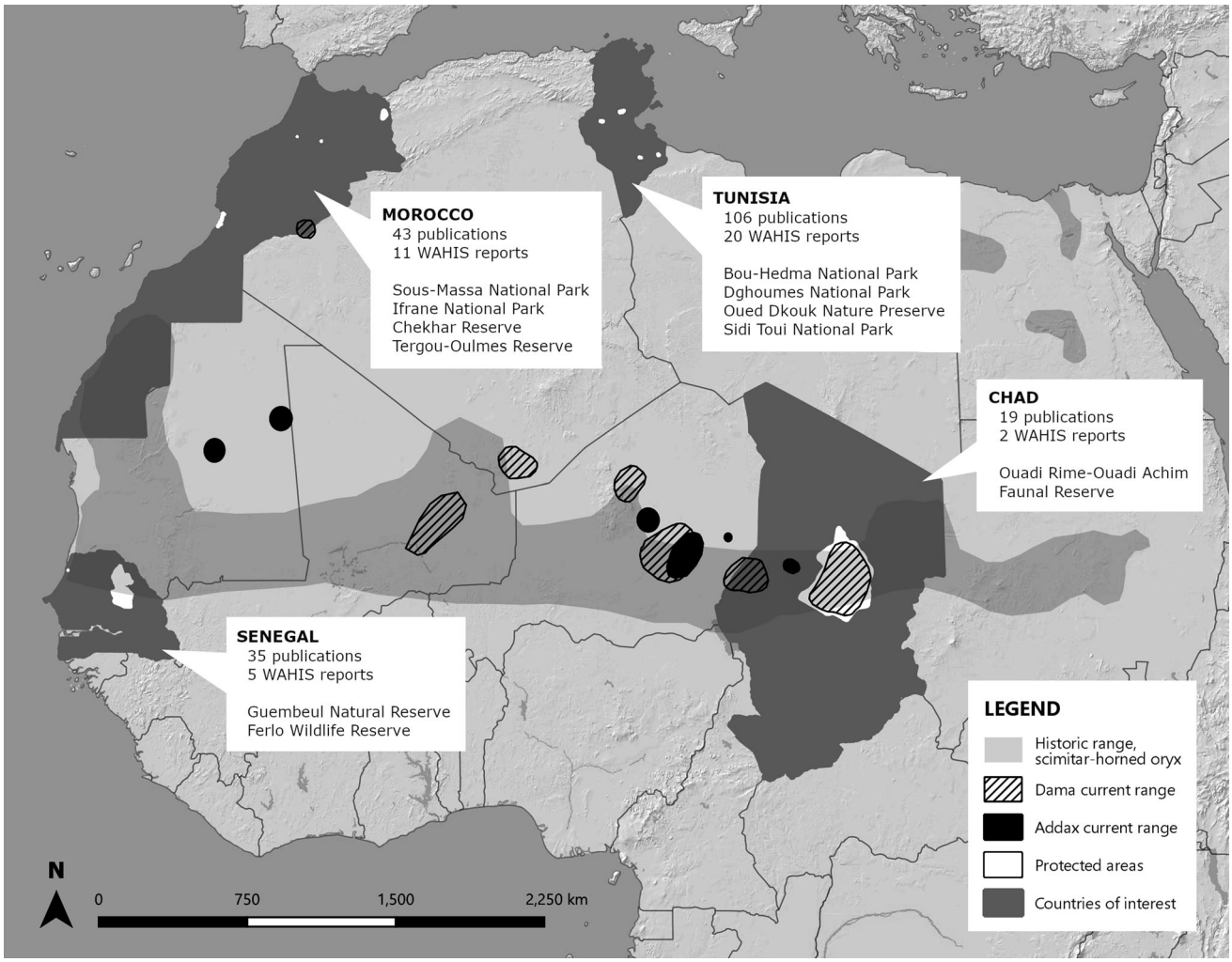


Figure 2. Countries of interest (dark gray), ranges of Sahelo-Saharan antelope (hashed, black, and light gray), and protected areas with long-term reintroduction programs (white). Results from literature review and names of protected areas shown in callouts. Data sources: GADM 4.1, World Database on Protected Areas, IUCN RedList of Threatened Species, ESRI basemap. Map created using QGIS 3.28.2 and GIMP 2.10.28.

mans generally become infected after direct or indirect contact with infected urine, and occupational exposure has been recognized as a significant risk factor (Levett, 2001). Also identified in our review were parasitic flukes of economic importance and zoonotic potential (*Fasciola gigantica*, *F. hepatica*, and to a lesser extent, *Dicrocoelium hospes*); *F. gigantica* and *D. hospes* in particular have been found in wild African ungulates (WOAH, 2021; Bengis et al., 2023). This review also identified some infectious agents that are non-pathogenic in our taxa of interest but carry significant health consequences for humans—including Crimean-Congo hemorrhagic fever virus (WOAH, 2021). Finally, a recent Chadian study estimated low brucellosis seroprevalence, but higher Q fever and RVF seroprevalence, in rural agro-pastoralist human communities (0.2%; 49.1%; and 28.1%, respectively) and their livestock (0.3%; 12.8%; and 10.2%, respectively). This study also found a positive association between human and animal RVF seropositivity, emphasizing the link between human and livestock health (Özcelik et al., 2023).

Perhaps unexpectedly, our literature review did not uncover reports of certain infectious diseases known to occur within the broader study region. For example, contagious caprine pleuropneumonia (CCPP), caused by *Mycoplasma capricolum* subsp. *capripneumoniae* (Mccp), is present in the Sahel, though its exact prevalence and distribution in Central and North Africa is uncertain (Manso-Silván and Thiaucourt, 2019). Due to its high morbidity and mortality rates and devastating effects on goats, it is listed as a WOAH-notifiable disease (WOAH, 2021). There are increasing reports of susceptibility and even high mortality rates in some wild ruminant species, including

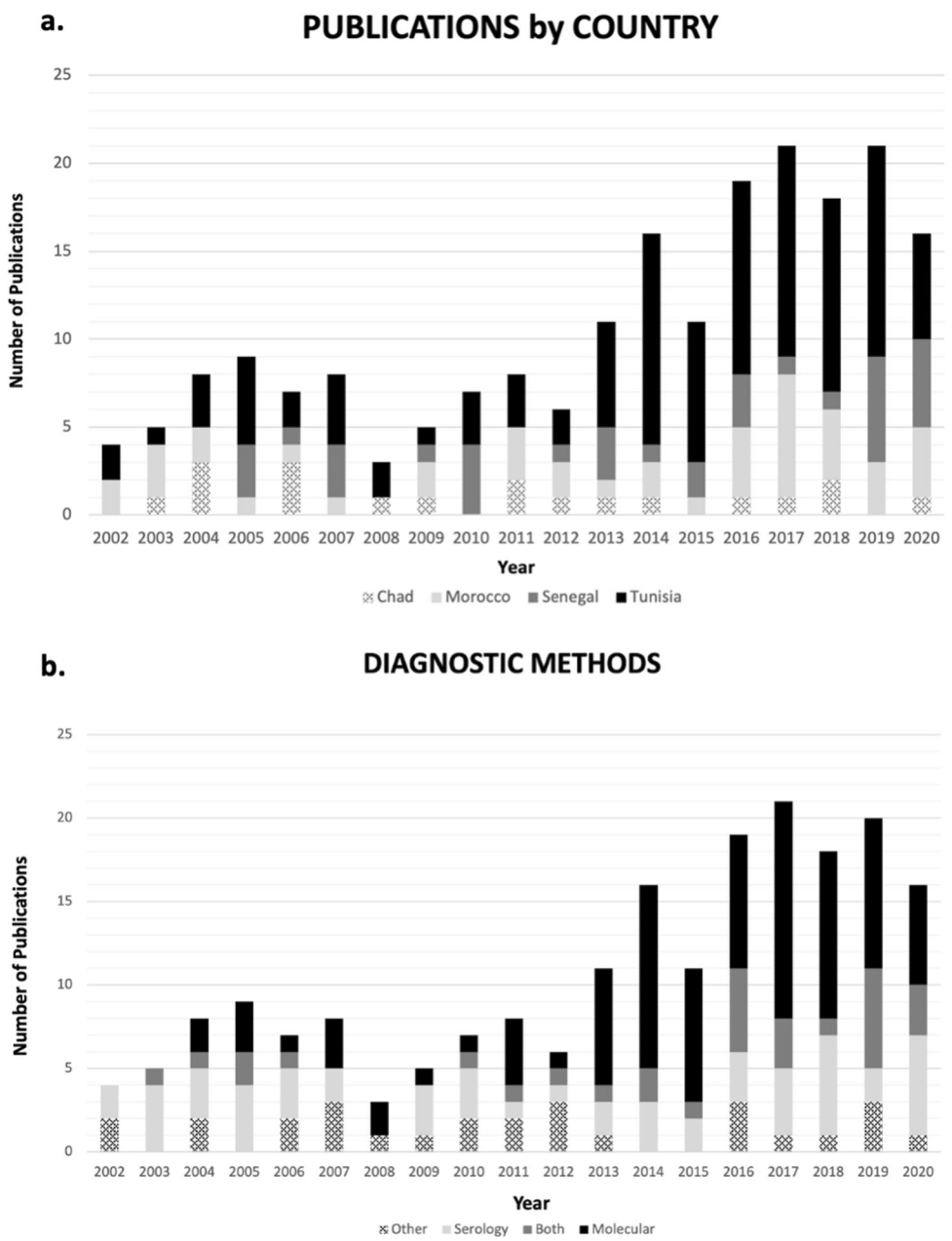


Figure 3. **a** Livestock infectious disease publications from each country of interest (Chad, Morocco, Senegal, and Tunisia) over time, 2002–2020. **b** Trends in serological and molecular diagnostic modalities in publications in countries of interest over time, 2002–2020.

gazelles (Lignereux et al., 2018). While hippotragine antelope were not initially thought to be susceptible, there has been at least one report of a fatal case in a captive Arabian oryx (Chaber et al., 2014). Conditions in the Sahel may contribute to Mccp transmission because animals often aggregate at limited watering, grazing, and shady sites (Manso-Silván and Thiacourt, 2019). In contrast, our WAHIS database search found reports of contagious bovine pleuropneumonia (CBPP), a similarly impactful disease for cattle, but with a more restricted host range that does not extend to gazelles or oryx (WOAH, 2021).

There are several possible explanations for observed mismatches between reports retrieved by our literature review and pathogens generally understood to occur across our region of interest. This review returned previous reports within temporal and geographic bounds based on active conservation initiatives for target species in immediate danger of extinction, not a complete timeline of disease events. As in any literature review, potential limi-

Table 7. Reports of Disease Events Retrieved From WOAH-WAHIS Database, 2005–2020.

Infectious Agent	Yr Event Start	Yr Reported	Host Species
CHAD			
Ehrlichia ruminantium	2013	2013	C
Rift Valley fever virus (RVFV)	2018	2019	W[1]
MOROCCO			
Bluetongue virus (BTV)	2006	2006	Sh
Bluetongue virus (BTV)	2009	2009	Sh
Bluetongue virus (BTV)	2009	2009	Sh
Bluetongue virus (BTV)	2010	2010	Sh
Bluetongue virus (BTV)	2010	2010	Sh
Epizootic hemorrhagic disease virus (EHDV)	2006	2006	Ct
Foot-and-mouth disease virus (FMDV)	2015	2015	Ct
Foot-and-mouth disease virus (FMDV)	2019	2019	Ct, G, Sh
Small ruminant morbillivirus (PPR virus)	2008	2008	G, Sh
Small ruminant morbillivirus (PPR virus)	2015	2015	G, Sh
Small ruminant morbillivirus (PPR virus)	2020	2020	G
SENEGAL			
Mycoplasma mycoides subsp. *mycoides*	2012	2012	Ct
Mycoplasma mycoides subsp. *mycoides*	2013	2013	Ct
Mycoplasma mycoides subsp. *mycoides*	2014	2014	Ct
Rift Valley fever virus (RVFV)	2013	2013	Ct, G, W[2]
Rift Valley fever virus (RVFV)	2020	2020	W[2,3]
TUNISIA			
Bluetongue virus (BTV)	2006	2006	Sh
Bluetongue virus (BTV)	2007	2007	Sh
Bluetongue virus (BTV)	2009	2009	Sh
Bluetongue virus (BTV)	2009	2009	Sh
Bluetongue virus (BTV)	2011	2011	Sh
Bluetongue virus (BTV)	2016	2016	Sh
Bluetongue virus (BTV)	2018	2018	Sh
Chlamydia abortus	2011	2012	G
Chlamydia abortus	2012	2012	Sh
Coxiella burnetii	2011	2012	G
Epizootic hemorrhagic disease virus (EHDV)	2006	2006	Ct
Epizootic hemorrhagic disease virus (EHDV)	2015	2015	Ct
Foot-and-mouth disease virus (FMDV)	2014	2014	Ct, G, Sh
Foot-and-mouth disease virus (FMDV)	2017	2017	Ct
Foot-and-mouth disease virus (FMDV)	2018	2018	Ct, Sh
Small ruminant morbillivirus (PPR virus)	2011	2011	G, Sh

Table 7. continued

Infectious Agent	Yr Event Start	Yr Reported	Host Species
Small ruminant morbillivirus (PPR virus)	2012	2012	Sh
Small ruminant morbillivirus (PPR virus)	2012	2012	G, Sh
Small ruminant morbillivirus (PPR virus)	2016	2016	G, Sh
Prion PrPSc (Scrapie)	2020	2020	Sh

C camels, *Ct* cattle, *G* goats, *Sh* sheep, *W* wildlife.
[1]Reported in scimitar-horned oryx (*Oryx dammah*).
[2]Reported in dorcas gazelle (*Gazella dorcas*).
[3]Reported in roan antelope (*Hippotragus equinus*).
*African horse sickness virus, African swine fever virus, equine influenza virus, myxomatosis virus, rabbit hemorrhagic disease virus, and West Nile virus also reported, but not in our species of interest.

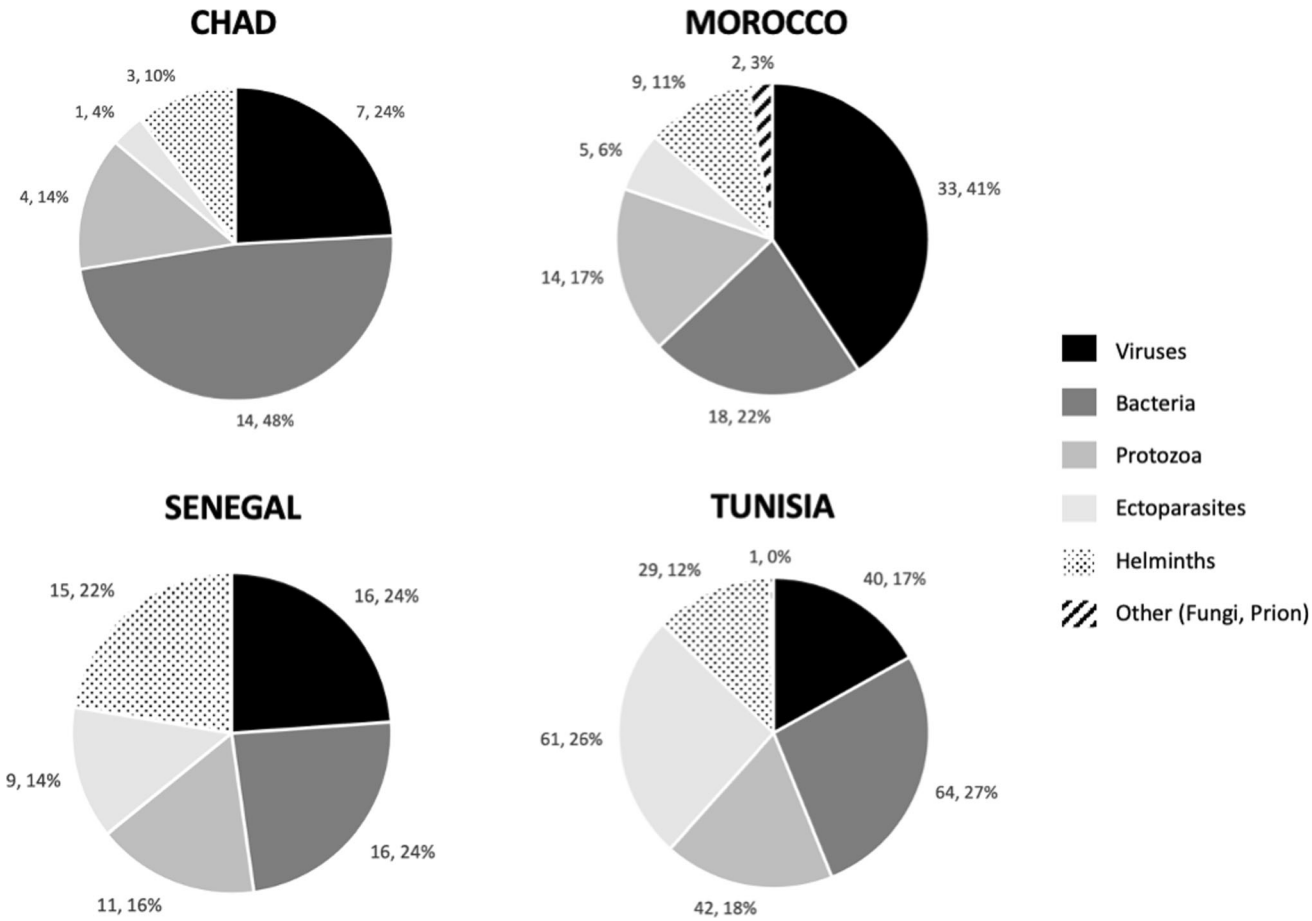


Figure 4. Frequency of reported infectious agents and parasites in domestic livestock and wild ruminants in countries of interest from peer-reviewed literature and WOAH-WAHIS database, by taxon/functional group. Results are presented over a total number of reports/mentions, not over the number of studies or records.

tations and biases in search terms and methods may have inappropriately eliminated some records. For example, French-language articles or regional journals may have been inadvertently excluded because they were not indexed in the searched databases. This review also does not reflect total reporting effort, as the searched databases may have inadequately captured conference proceedings, dissertations, book chapters, reports, and other publication types

Table 8. Most frequently reported infectious agents and parasites in domestic and wild ruminants by country of interest, 2002–2020

CHAD		MOROCCO		SENEGAL		TUNISIA	
Agent/ Parasite	# Reports	Agent/ Parasite	# Reports	Agent/ Parasite	# Reports	Agent/ Parasite	# Reports
Mycobacterium sp.	5	Bluetongue virus	7	RVF virus	7	*Hyalomma sp.*	25
Trypanosoma sp.	4	*Anaplasma sp.*	6	*Trypanosoma sp.*	6	*Rhipicephalus sp.*	19
RVF virus	3	*Toxoplasma gondii*	4	*Anaplasma sp.*	5	*Anaplasma sp.*	18
Bacillus anthracis	2	*Babesia sp.*	3	*Schistosoma sp.*	5	*Theileria sp.*	14
Brucella sp.	2	*Brucella sp.*	3	*Hyalomma sp.*	4	Bluetongue virus	10
Coxiella burnetti	2	*Chlamydia abortus*	3	*Mycoplasma mycoides*	4	*Babesia sp.*	8
Fasciola gigantica	2	*Hyalomma sp.*	3	*Rhipicephalus sp.*	4	*Chlamydia sp.*	8
FMD virus	2	*Leptospira sp.*	3	*Brucella sp.*	3	*Toxoplasma gondii*	8
		Theileira annulata	3	*Thysaniezia sp.*	3	*Echinococcus sp.*	7
	7					*Phlebotomus sp.*	
						PPR virus	7

Includes published peer-reviewed literature and records from the WOAH-WAHIS database. Results are presented over a total number of reports, not over the number of studies or WOAH-WAHIS records

Table 9. Most frequently reported infectious agents and parasites by species of interest (camels, cattle, sheep, goats) in Chad, Morocco, Senegal, and Tunisia, 2002–2020

CAMELS		CATTLE		SHEEP		GOATS	
Agent/ Parasite	# Reports	Agent/ Parasite	# Reports	Agent/ Parasite	# Reports	Agent/ Parasite	# Reports
Hyalomma sp.	18	*Anaplasma sp.*	14	*Rhipicephalus sp.*	20	PPR virus	14
Anaplasma sp.	6	*Hyalomma sp.*	11	Bluetongue virus	16	*Anaplasma sp.*	11
Echinococcus sp.	6	*Mycobacterium sp.*	11	*Anaplasma sp.*	14	*Rhipicephalus sp.*	9
Rickettsia sp.	5	*Theileria sp.*	11	PPR virus	12	RVF virus	9
Phlebotomus sp.	4	*Brucella sp.*	10	*Toxoplasma gondii*	12	*Phlebotomus sp.*	7
Trypanosoma sp.	4	*Trypanosoma sp.*	8	*Chlamydia sp.*	8	*Toxoplasma gondii*	7
Bluetongue virus	3	*Babesia sp.*	7	RVF virus	8	*Chlamydia sp.*	6
MERS coronavirus	3	FMD virus	7	*Phlebotomus sp.*	7	*Schistosoma sp.*	5
		Phlebotomus sp.	7	*Echinococcus sp.*	6		
		Rhipicephalus sp.	7	*Hyalomma sp.*	6		

(Saidi, 2021). Additional mismatches could also stem from sporadic reporting of disease events, or a lack of recent research interest into infectious diseases known to be well-established in the region. Biases in reporting or publication may also be due to the prioritization of food-borne or other diseases of human health significance (e.g., cystic echinococcosis, schistosomiasis), or pathogens that specifically impact livestock production (e.g., *Haemonchus*). Finally, regional sociopolitical factors like political instability and conflict may also impact the ability to conduct and publish in situ research.

Not all infectious agents identified in this review pose a serious health threat to our taxa of interest. For many of these agents, direct evidence of clinical effects in gazelles or hippotragines is lacking. Additionally, many of the infectious agents identified are known to be fairly benign in their hosts and likely represent incidental or subclinical infections. For example, *Theileria annulata* and *T. lestoquardi* cause clinical disease in cattle and sheep, respectively, but *T. ovis* and *T. buffeli* are largely non-pathogenic (WOAH, 2021). Similarly, *Anaplasma marginale* is the principal cause of clinical bovine anaplasmosis, whereas other species

identified in the review (e.g., *A. bovis*) are generally more benign in our ruminant taxa of interest (WOAH, 2021). Reports using serologic methods also demonstrate evidence of exposure, but not necessarily active infection or disease. Finally, the countries included in the review represent distinct locales, and local variants or strains may vary in pathogenicity.

Conclusion

This literature review catalogs infectious agents relevant to livestock and endangered Sahelo-Saharan antelope across Chad, Morocco, Senegal, and Tunisia. While surveillance in domestic livestock can help inform conservation and reintroduction programs, more targeted studies of pathogen shedding and contact rates at livestock-wildlife interfaces are needed to advance our understanding of cross-species disease transmission dynamics. Investigations into wildlife and domestic livestock movements and interactions in shared Sahelo-Saharan landscapes would clarify the role of free-ranging ungulates in the epidemiology of livestock diseases, and vice versa. With climate change projected to substantially affect arid regions of Africa, these interfaces may become increasingly important sites for disease transmission and necessitate continual reassessment of infectious disease risks for Sahelo-Saharan antelope.

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